Milestone 1: Multi-Agent Company Profiling Pipeline

Produce a reliable, consolidated 10-field company profile, a skill-set table, and a hiring-process JSON — all by running four Python scripts manually and reading the output. No database writes yet; that comes in Milestone 2.

4

Python Scripts

Run manually end-to-end

3

LLM Providers

For triangulated research

10

Profile Fields

Per Golden Record JSON

12

Skill Areas

With level + proficiency code

The Four Agents

#	Agent	Input → Output	Providers
1	Researcher	Company name → 10-field JSON × 3 (one per provider)	Groq, Google, Cerebras
2	Consolidator	3 Researcher JSONs → 1 cherry-picked "Golden Record" JSON	OpenRouter
3	Skill Set	Company name → JSON with 12 skill areas × level + proficiency code	Groq
4	Hiring Process	Company name → JSON of roles, rounds, and typical interview questions	Groq

Architecture, Rationale & Success Checklist

Why Three Providers for the Researcher?

Different LLMs hallucinate differently. Running the same prompt through 3 providers gives us triangulation — the Consolidator then picks the best value per field. This keeps the demo honest: a single provider alone would hide the quality gap visible across vendors.

Why a 4th Provider for the Consolidator?

Cross-provider independence. If we consolidated using one of the same three providers, that provider's answers would self-bias. OpenRouter gives us an easy fallback catalogue and a single key covering many models.

Manual Run Flow

Nothing writes to the database yet — that happens in Milestone 2's ETL agent.

Success Checklist

1

End-to-End Script Execution

All four scripts run end-to-end for "Fractal" without errors or interruptions.

2

Pydantic Schema Validation

Output JSON passes the Pydantic schema defined alongside each agent — no malformed fields.

3

No Placeholder Text

No `"TBD"`, `"Lorem ipsum"`, or stub values survive the validator at any stage.

4

Retry Loop Triggered

The retry loop triggers visibly at least once during the demo — we will force it to prove resilience.